A Proactive Machine Learning Framework for Women's Night-Time Safety Using Crime Forecasting and Anomaly Detection

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***Abstract*— Night-time urban safety remains a critical challenge, often hindered by reactive policing. This research proposes a proactive, data-driven framework for women's safety by integrating crime forecasting with intelligent pathfinding. By modeling urban environments as weighted graphs based on historical crime data, we identify high-risk "hotspots." We evaluate and compare three optimization strategies: Dijkstra's algorithm and A* for point-to-point safe routing, and Genetic Algorithms (GA) for multi-zone patrol optimization. Results show the framework effectively balances travel efficiency with maximum risk coverage. This transition from reactive to predictive patrolling provides a strategic resource-allocation tool to enhance the security of vulnerable areas in smart city environments..**

Keywords— ***Crime Forecasting, Women's Safety, Anomaly Detection, Genetic Algorithms, Patrol Route Optimization, Dijkstra's Algorithm, A* Search, Risk Assessment, Proactive Policing, Smart City Security***

I. Introduction

Women's safety during night-time is still a major concern in urban areas, where traditional patrolling incidents befor e they happen. Even with growing awareness and technology, certain zones remain high-risk due to factors like poor street lighting, historical crime data, and delayed responses. In our research, we developed an **intelligent, risk-based patrolling framework** that focuses on areas with higher potential for incidents. Using historical crime records and risk indices, we identified **high-risk zones** and designed optimized patrol routes so that teams can reach these areas efficiently. We implemented three algorithms—**Dijkstra's shortest path, A*, and Genetic Algorithms (GA)**—to plan patrols. While Dijkstra and A* help in finding the safest routes to individual high-risk zones, GA is used to create **optimized routes covering multiple high-risk zones**, balancing both risk and distance.

Our framework moves beyond reactive safety measures and provides a **proactive patrolling system**. By combining riskaware routing with algorithmic optimization, we can help patrol teams cover the most vulnerable areas effectively, improving women's safety at night and supporting datadriven decision-making for city authorities.

II . .Motivation

Despite the numerous efforts to improve women's safety in urban areas, incidents still occur, especially during nighttime when response times are slower and certain zones remain more vulnerable. We observed that traditional patrolling systems are often **reactive**, responding after incidents happen, rather than **proactively preventing them**. This motivated us to explore a **datadriven and algorithmic approach** for risk-based patrolling. By analyzing historical crime data and assigning risk levels to different zones, we can identify the areas that need immediate attention. Our goal is to design an **intelligent system that helps patrolling teams reach high-risk zones efficiently**, ensuring that resources are allocated where they are needed most. Through this research, we aim to **shift from reactive to proactive safety measures**, providing

III. .LITERATURE REVIEW

1.Jiang et al. (2022) proposed a **multi-objective genetic algorithm (DP-MOGA)** for street patrol routing in Zhengzhou, China. They classified patrol segments by incident frequency and optimized routes to minimize average response time and patrol personnel requirements. Their results demonstrated that evolutionary optimization outperforms traditional heuristic routing methods under variable traffic and risk conditions.

2.Dewinter, Vandeviver, Vander Beken, and Witlox (2020) conducted a comprehensive review of the **Police Patrol Routing Problem (PPRP)**, linking patrol routing with dynamic vehicle routing problems. They discussed a range of solution techniques including genetic algorithms, linear and

stochastic optimization, and hybrid methods. This review highlighted the challenges of patrol planning under uncertain crime patterns and pointed to gaps in systematic, risk-aware routing strategies

3.Rumi, Qin, and Salim (2020) formulated patrol planning as a joint learning and optimization problem using **multisource data (crime records, check-ins, incident responses)** combined with meta-heuristic algorithms such as genetic algorithms and Cuckoo Search. Their work showed that integrating multiple data sources can significantly improve patrol routing efficiency and highrisk area coverage compared to baseline methods.

4.Kim, Kan, and Yi (2019) developed a **big-data-driven patrol route optimization framework** that uses real-time crime and traffic data with genetic algorithm optimization to extract hotspots and generate patrol sequences with minimal response times. Their research illustrated the benefit of combining real-time analytics with evolutionary optimization to enhance patrol responsiveness and risk coverage.

IV . OBJECTIVE

The main objective of this research is to **develop an** • **intelligent, risk-aware patrolling framework** that ensures high-risk zones are reached efficiently by patrol teams, • thereby enhancing women's safety during night-time in urban areas. The specific objectives are as follows: •

1. To **analyze historical crime data and risk indices** to identify high-risk zones within the urban environment.
2. To **design and implement algorithmic strategies**, • including Dijkstra, A*, and Genetic Algorithms, for planning efficient patrol routes targeting high-risk areas.
3. To **prioritize patrol coverage** based on the severity of risk, distance, and accessibility, ensuring the most vulnerable zones are patrolled first.
4. To **optimize patrol routes** for multiple zones using Genetic Algorithms, balancing risk coverage, response time, and route efficiency.
5. To provide a **proactive, data-driven framework** for law enforcement that supports decision-making, resource allocation, and real-time patrolling strategies to enhance women's safety.

I. V. METHODOLOGY

The methodology of this research focuses on designing a **risk-aware patrolling framework** for women's safety at night.

1. Data Collection and Preprocessing: Historical crime data is collected, including incident type, location, and time. Each zone is assigned a **risk score**, and the data is structured into a **graph**, where zones are nodes and streets are weighted edges based on distance and risk.

2. Identification of High-Risk Zones: Zones with higher risk scores are identified as **priority areas** for patrolling.

3. Algorithmic Path Planning: Three algorithms are used: **Dijkstra** to find the shortest safe path to individual highrisk zones, **A*** for faster heuristic-based pathfinding, and **Genetic Algorithm (GA)** to optimize patrol routes covering multiple high-risk zones efficiently.

4. Risk-Aware Route Optimization: GA generates patrol routes that prioritize high-risk zones while minimizing total risk and travel cost. Fitness functions consider both **risk scores and edge weights**.

5. Evaluation and Visualization: Routes from Dijkstra, A*, and GA are compared based on total risk, distance, and high-risk zone coverage. Routes are visualized on the graph to assess patrol efficiency.

6.Implementation and Testing: The methodology is implemented using **Python**, with **NetworkX** for graph modeling and **DEAP** for GA. Multiple simulations verify that patrols reach high-risk areas effectively.

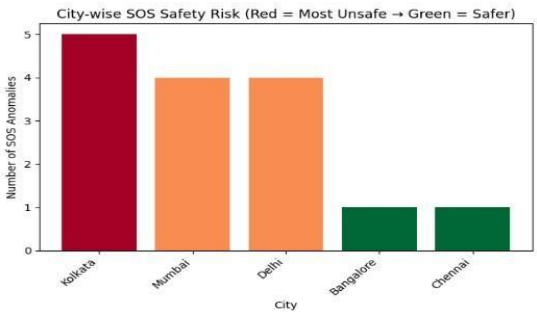
VI. EXPECTED OUTCOME

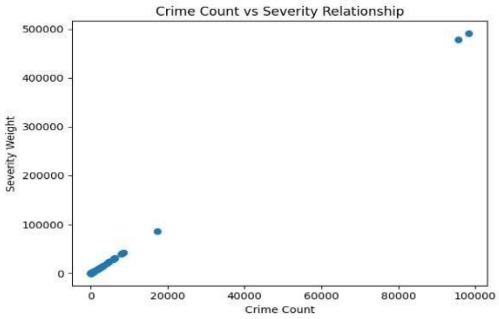
Identification of **high-risk zones** in the city using historical crime data and risk scoring.

Generation of **efficient patrol routes** using Dijkstra and A* algorithms for individual high-risk zones.

Optimization of **multi-zone patrol routes** using Genetic Algorithm to ensure maximum coverage of high-risk areas with minimum travel risk.

A **data-driven, risk-aware patrolling framework** that helps law enforcement proactively reach vulnerable zones and improve women's safety during night-time.





evolutionary optimization as an effective approach for solving complex multi-objective problems.

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.VII. Challenges Faced

Data Limitations: Difficulty in obtaining accurate and complete crime records for proper risk scoring and analysis.

1.Algorithm Implementation: Designing and tuning the Genetic Algorithm to balance risk, distance, and coverage, while avoiding errors during crossover and mutation.

2.Urban Network Modeling: Representing real-world streets and zones in a graph format and ensuring patrol routes prioritize high-risk areas efficiently.

VIII. .ALOGORITHM SUMARY TABLE

Algorithm	Purpose	Application
Dijkstra	Find shortest path	Routes to single high-risk zones
A*	Heuristic based shortest path	Efficient path to high-risk zones
Genetic Algorithm	Optimize multi-zone routes	Patrol multiple high-risk zones efficiently

IX. Reference

I. Shortest path algorithms are widely used for routing problems, and a foundational contribution was made by E. W. Dijkstra in *"A Note on Two Problems in Connexion with Graphs"* published in **Numerische Mathematik**, 1959, where the shortest path algorithm was introduced and proven to guarantee optimal paths.

II. Hart, Nilsson, and Raphael proposed the A* search algorithm in their paper *"A Formal Basis for the Heuristic Determination of Minimum Cost Paths"* published in **IEEE Transactions on Systems Science and Cybernetics**, 1968, demonstrating how heuristic guidance improves search efficiency over traditional methods.

III. Holland introduced the concept of Genetic Algorithms in *"Adaptation in Natural and Artificial Systems"*, University of Michigan Press, 1975, establishing

IV. Brantingham and Brantingham, in *"Environmental Criminology"* published in **Waveland Press**, 1991, analyzed spatial crime patterns and emphasized the importance of geographic and environmental factors in crime prevention and policing strategies.